

MODEL RUN TRANSCRIPT

Generated: 2026-08-10T10:07:29

Source: data/json/noise_experiments/v2/noise2i_memprimary_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary/noise2i_memprimary_game_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 4
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 4
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$5.5; payoffs Agent_1 (Aster)=\$7.5, Agent_2 (Briar)=\$3.5; noise/visible: sent visible=\$3.89, received visible=\$11.67, returned visible=\$5.21

Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.5,

"Agent_2": 3.5

}

Visible balances after round: {

"Agent_1": 7.21,

"Agent_2": 6.17

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:04:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance the potential gains from cooperation against the risk that the receiver might not return much.

Since this is a multi-round game, establishing some trust could benefit both of us in the long run. If I send money and the receiver returns a fair amount, we can both end up better off than keeping everything.

I'll send a moderate amount as an initial trust signal - not so much that I'm left with nothing if they don't reciprocate, but enough to show willingness to cooperate.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There may be communication noise (amounts may differ slightly)

This is a classic trust game scenario. The economically rational strategy depends on several factors:

1. If I assume the receiver is purely self-interested and this is a one-shot game, they would return \$0, so I should send \$0
2. However, if there are multiple rounds and I can build trust, it might be beneficial to send money
3. The communication noise adds uncertainty - amounts may be misreported

Since this appears to be a multi-round game (the prompt mentions "multiple rounds"), there's potential for reciprocity and trust-building.

A reasonable first-round strategy might be to:

- Send a moderate amount to test the receiver's trustworthiness
- Not send everything (too risky)
- Not send nothing (no opportunity to build trust)

Sending \$2-3 seems reasonable as an initial trust signal. This would give the receiver \$6-9 to work with, and if they return half or more, both parties benefit.

Let me send \$3 as a moderate trust signal. This gives the receiver \$9, and if they return even \$3-4, we both gain from cooperation.

Wait, let me reconsider the payoff structure:

- If I send \$3: I have \$2 left, receiver gets \$9
- If receiver returns \$4: I end with \$6 total (\$2 + \$4), receiver ends with \$5 (\$9 - \$4)
- Both parties are better off than the initial \$5 each

I'll go with \$3 as a balanced initial trust signal.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 548,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:04:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
```

```

"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to The sender? (0-11.67)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)

- They currently have \$1.11 left
- I received \$11.67 after the 3x multiplication
- Total money: \$12.78

The sender took a risk by trusting me with most of their endowment. To reciprocate this trust fairly, I should return enough so we both benefit from the cooperation.

If I return approximately \$5.50:

- Sender ends with: $\$1.11 + \$5.50 = \$6.61$
- I end with: $\$11.67 - \$5.50 = \$6.17$
- Both of us gain compared to the initial \$5

This rewards their trust while maintaining fairness.

```
{'return': 5.50}
```

Reasoning:

I'm playing as the RECEIVER in this trust game. Let me understand the situation:

1. The sender sent \$3.89 (which is 77.8% of their \$5 endowment)
2. This was multiplied by 3x, so I received \$11.67
3. I need to decide how much to return (between \$0 and \$11.67)

Let me think about this strategically:

- The sender showed significant trust by sending 77.8% of their endowment
- If I return nothing, I keep \$11.67 and the sender gets only $\$5 - \$3.89 = \$1.11$
- If I return a fair amount, we can both benefit

Let me calculate what would be fair:

- The sender started with \$5
- They sent \$3.89, keeping \$1.11
- I received \$11.67

A common reciprocity approach would be to:

1. Split the gains equally, or
2. Return enough so the sender doesn't regret trusting me

If we split equally:

- Total money in the system: $\$5$ (sender's initial) + $\$11.67$ (what I received) = $\$16.67$
- Wait, that's not right. Let me recalculate.

Actually:

- Sender's initial endowment: \$5
- Sender sent: \$3.89
- I received: $\$3.89 \times 3 = \11.67
- Sender currently has: $\$5 - \$3.89 = \$1.11$
- I currently have: \$11.67

Total in system: $\$1.11 + \$11.67 = \$12.78$

If we split the total equally:

- Each person gets: $\$12.78 / 2 = \6.39
- Sender has \$1.11, needs \$6.39, so I should return: $\$6.39 - \$1.11 = \$5.28$
- This would give us equal outcomes at \$6.39 each

I could also return roughly half of what I received (\$5.84) to reward their trust, which would leave them with \$6.95 while I keep \$5.83. Or I could simply return what they sent (\$3.89) so they break even on the transaction itself.

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

```

dyad_1: Agent_2 (Briar) sent $3.5; received $10.5; Agent_1 (Aster) returned $0; payoffs Agent_2
(Briar)=$1.5, Agent_1 (Aster)=$10.5; noise/visible: sent visible=$0, received visible=$0, returned
visible=$0.22
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 10.5, "min": 0, "raw":
0.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 18.0,
  "Agent_2": 5.0
}
Visible balances after round: {
  "Agent_1": 7.21,
  "Agent_2": 7.89
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-10T10:04:38
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to The sender? (0-11.67)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- They currently have \$1.11 left
- I received \$11.67 after the 3x multiplication
- Total money: \$12.78

The sender took a risk by trusting me with most of their endowment. To reciprocate this trust fairly, I should return enough so we both benefit from the cooperation.

If I return approximately \$5.50:

- Sender ends with: $\$1.11 + \$5.50 = \$6.61$
- I end with: $\$11.67 - \$5.50 = \$6.17$
- Both of us gain compared to the initial \$5

This rewards their trust while maintaining fairness.

{'return': 5.50}

reasoning:

I'm playing as the RECEIVER in this trust game. Let me understand the situation:

1. The sender sent \$3.89 (which is 77.8% of their \$5 endowment)
2. This was multiplied by 3x, so I received \$11.67
3. I need to decide how much to return (between \$0 and \$11.67)

Let me think about this strategically:

- The sender showed significant trust by sending 77.8% of their endowment
- If I return nothing, I keep \$11.67 and the sender gets only $\$5 - \$3.89 = \$1.11$
- If I return a fair amount, we can both benefit

Let me calculate what would be fair:

- The sender started with \$5
- They sent \$3.89, keeping \$1.11
- I received \$11.67

A common reciprocity approach would be to:

1. Split the gains equally, or
2. Return enough so the sender doesn't regret trusting me

If we split equally:

- Total money in the system: $\$5$ (sender's initial) + $\$11.67$ (what I received) = $\$16.67$
- Wait, that's not right. Let me recalculate.

Actually:

- Sender's initial endowment: \$5
- Sender sent: \$3.89
- I received: $\$3.89 \times 3 = \11.67
- Sender currently has: $\$5 - \$3.89 = \$1.11$
- I currently have: \$11.67

Total in system: $\$1.11 + \$11.67 = \$12.78$

If we split the total equally:

- Each person gets: $\$12.78 / 2 = \6.39
- Sender has \$1.11, needs \$6.39, so I should return: $\$6.39 - \$1.11 = \$5.28$
- This would give us equal outcomes at \$6.39 each

I could also return roughly half of what I received (\$5.84) to reward their trust, which would leave them with \$6.95 while I keep \$5.83. Or I could simply return what they sent (\$3.89) so they break even on the transaction itself.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$6.17.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at this situation:

In Round 1, the sender showed considerable trust by sending me 77.8% of their endowment, and I reciprocated with a fair return. This round, I'm in the sender position.

To maintain the cooperative pattern and signal trust to this receiver, I should send a substantial amount. Sending around 70% demonstrates:

- Willingness to trust and cooperate
- Hope for mutual benefit
- Reciprocation of the trust shown to me

If I send \$3.50 (70%):

- Receiver gets \$10.50
- If they reciprocate fairly, both of us can benefit
- This maintains the cooperative spirit from Round 1

{'send': 3.50}

Reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- The sender sent me \$3.89 (77.8% of their endowment)
- I returned \$5.50
- My current total earnings are \$6.17

This suggests the sender was quite trusting, and I reciprocated with a fair return that benefited both of us.

Now I'm in the sender position. I need to decide how much to send.

Key considerations:

1. The sender in Round 1 showed high trust (77.8%)
2. I reciprocated fairly
3. This might establish a pattern of cooperation
4. However, this is a new round with potentially a different partner

Given the experimental context and the note about communication noise, I should consider:

- Sending a reasonable amount to signal trust
- Not necessarily the exact same percentage as Round 1 sender
- Being somewhat trusting but also cautious since I don't know if this is the same partner

If I send \$3:

- They receive \$9
- If they return proportionally (around 50%), I'd get back ~\$4.50
- I'd end with $\$2 + \$4.50 = \$6.50$
- They'd keep ~\$4.50

If I send \$4:
- They receive \$12
- If they return proportionally, I'd get back ~\$6
- I'd end with \$1 + \$6 = \$7
- They'd keep ~\$6

If I send \$2:
- They receive \$6
- If they return proportionally, I'd get back ~\$3
- I'd end with \$3 + \$3 = \$6
- They'd keep ~\$3

I think sending around \$3 (60%) shows trust and willingness to cooperate, similar to the previous round, but is slightly less risky than the 77.8% sent to me. This balances trust with caution.

```
Usage: {  
  "input_tokens": 630,  
  "output_tokens": 703,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:04:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential gains from cooperation against the risk that the receiver might not return much.

Since this is a multi-round game, establishing some trust could benefit both of us in the long run. If I send money and the receiver returns a fair amount, we can both end up better off than keeping everything.

I'll send a moderate amount as an initial trust signal - not so much that I'm left with nothing if they don't reciprocate, but enough to show willingness to cooperate.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There may be communication noise (amounts may differ slightly)

This is a classic trust game scenario. The economically rational strategy depends on several factors:

1. If I assume the receiver is purely self-interested and this is a one-shot game, they would return \$0, so I should send \$0
2. However, if there are multiple rounds and I can build trust, it might be beneficial to send money
3. The communication noise adds uncertainty - amounts may be misreported

Since this appears to be a multi-round game (the prompt mentions "multiple rounds"), there's potential for reciprocity and trust-building.

A reasonable first-round strategy might be to:

- Send a moderate amount to test the receiver's trustworthiness
- Not send everything (too risky)
- Not send nothing (no opportunity to build trust)

Sending \$2-3 seems reasonable as an initial trust signal. This would give the receiver \$6-9 to work with, and if they return half or more, both parties benefit.

Let me send \$3 as a moderate trust signal. This gives the receiver \$9, and if they return even \$3-4, we both gain from cooperation.

Wait, let me reconsider the payoff structure:

- If I send \$3: I have \$2 left, receiver gets \$9
- If receiver returns \$4: I end with \$6 total (\$2 + \$4), receiver ends with \$5 (\$9 - \$4)
- Both parties are better off than the initial \$5 each

I'll go with \$3 as a balanced initial trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 548,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.21.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 ($0 \times 3 = \$0$). There's nothing for me to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me understand this situation:

- I'm now the RECEIVER
- The sender sent me \$0
- This means I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (which means I can only return \$0)
- My total earnings across all rounds are \$7.21

Since I received \$0, there's nothing to return. The sender chose not to send me anything this round.

The response format should be: {'return': <amount>}

Since I received \$0, I can only return \$0.

```
Usage: {
  "input_tokens": 550,
  "output_tokens": 188,
  "reasoning_tokens": 0
}
```

=====

ROUND 3

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$4; received \$12; Agent_2 (Briar) returned \$0.8; payoffs Agent_1 (Aster)=\$1.8, Agent_2 (Briar)=\$11.2; noise/visible: sent visible=\$0.7, received visible=\$2.1, returned visible=\$1.5

Action validation: {"returned": {"amount": 0.8, "clamped": false, "max": 12.0, "min": 0, "raw": 0.8}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

```
"Agent_1": 19.8,
"Agent_2": 16.2
}
```

Visible balances after round: {

```
"Agent_1": 9.71,
"Agent_2": 9.19
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:04:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential gains from cooperation against the risk that the receiver might not return much.

Since this is a multi-round game, establishing some trust could benefit both of us in the long run. If I send money and the receiver returns a fair amount, we can both end up better off than keeping everything.

I'll send a moderate amount as an initial trust signal - not so much that I'm left with nothing if they don't reciprocate, but enough to show willingness to cooperate.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There may be communication noise (amounts may differ slightly)

This is a classic trust game scenario. The economically rational strategy depends on several factors:

1. If I assume the receiver is purely self-interested and this is a one-shot game, they would return \$0, so I should send \$0
2. However, if there are multiple rounds and I can build trust, it might be beneficial to send money
3. The communication noise adds uncertainty - amounts may be misreported

Since this appears to be a multi-round game (the prompt mentions "multiple rounds"), there's potential for reciprocity and trust-building.

A reasonable first-round strategy might be to:

- Send a moderate amount to test the receiver's trustworthiness
- Not send everything (too risky)
- Not send nothing (no opportunity to build trust)

Sending \$2-3 seems reasonable as an initial trust signal. This would give the receiver \$6-9 to work with, and if they return half or more, both parties benefit.

Let me send \$3 as a moderate trust signal. This gives the receiver \$9, and if they return even \$3-4, we both gain from cooperation.

Wait, let me reconsider the payoff structure:

- If I send \$3: I have \$2 left, receiver gets \$9
- If receiver returns \$4: I end with \$6 total (\$2 + \$4), receiver ends with \$5 (\$9 - \$4)
- Both parties are better off than the initial \$5 each

I'll go with \$3 as a balanced initial trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 548,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.21.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant**content:**

Since the sender sent me \$0, I received \$0 ($0 \times 3 = \$0$). There's nothing for me to return.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand this situation:

- I'm now the RECEIVER
- The sender sent me \$0
- This means I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (which means I can only return \$0)
- My total earnings across all rounds are \$7.21

Since I received \$0, there's nothing to return. The sender chose not to send me anything this round.

The response format should be: {'return': <amount>}

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 188,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.21.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my position:

In Round 1, I sent \$3 and based on my total earnings of \$7.21, the receiver returned a fairly generous amount. This suggests they were cooperative.

In Round 2, I was the receiver and got nothing (the sender sent \$0).

Now in Round 3, I'm a sender again. Given that my previous trust was reasonably rewarded, I'll send a bit more this time to continue building cooperation.

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$7.21.

Let me calculate what happened:

- I started with \$5
- I sent \$3
- The receiver got \$9 (3×3)
- My earnings after Round 1 would be: $\$5 - \$3 + \text{returned amount} = \$2 + \text{returned amount}$

If my total earnings are \$7.21 after Round 1, that means:

$\$2 + \text{returned} = \7.21

$\text{returned} = \$5.21$

Wait, that doesn't seem right. Let me reconsider.

Actually, looking at Round 2:

- I was the RECEIVER
- The sender sent me \$0
- I received \$0 and returned \$0
- So I gained \$0 in Round 2

So my total earnings of \$7.21 should be from Round 1 only.

Round 1 calculation (I was SENDER):

- Started with \$5
- Sent \$3

- Receiver got \$9
- I kept \$2 initially
- Receiver returned some amount to me
- My Round 1 earnings = \$2 + returned

If total earnings = \$7.21 after Round 2, and Round 2 gave me \$0, then Round 1 gave me \$7.21.
 So: \$2 + returned = \$7.21
 returned = \$5.21

Hmm, that seems like the receiver returned \$5.21 out of the \$9 they received. That's actually quite generous (about 58% return rate).

Now for Round 3, I'm the SENDER again.

Given that:

1. In Round 1, I sent \$3 and the receiver returned a generous \$5.21
2. In Round 2, when I was receiver, the sender sent me \$0 (possibly a different person?)
3. The note mentions there may be communication noise, so amounts might differ

```
Usage: {
  "input_tokens": 641,
  "output_tokens": 687,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:05:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.89, that is 77.8% of their total endowment. You received \$11.67. How much do you return to The sender? (0-11.67)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.89 (77.8% of their endowment)
- They currently have \$1.11 left
- I received \$11.67 after the 3x multiplication
- Total money: \$12.78

The sender took a risk by trusting me with most of their endowment. To reciprocate this trust fairly, I should return enough so we both benefit from the cooperation.

If I return approximately \$5.50:

- Sender ends with: $\$1.11 + \$5.50 = \$6.61$
- I end with: $\$11.67 - \$5.50 = \$6.17$
- Both of us gain compared to the initial \$5

This rewards their trust while maintaining fairness.

{'return': 5.50}

reasoning:

I'm playing as the RECEIVER in this trust game. Let me understand the situation:

1. The sender sent \$3.89 (which is 77.8% of their \$5 endowment)
2. This was multiplied by 3x, so I received \$11.67
3. I need to decide how much to return (between \$0 and \$11.67)

Let me think about this strategically:

- The sender showed significant trust by sending 77.8% of their endowment
- If I return nothing, I keep \$11.67 and the sender gets only $\$5 - \$3.89 = \$1.11$
- If I return a fair amount, we can both benefit

Let me calculate what would be fair:

- The sender started with \$5
- They sent \$3.89, keeping \$1.11
- I received \$11.67

A common reciprocity approach would be to:

1. Split the gains equally, or
2. Return enough so the sender doesn't regret trusting me

If we split equally:

- Total money in the system: $\$5$ (sender's initial) + $\$11.67$ (what I received) = $\$16.67$
- Wait, that's not right. Let me recalculate.

Actually:

- Sender's initial endowment: \$5
- Sender sent: \$3.89
- I received: $\$3.89 \times 3 = \11.67
- Sender currently has: $\$5 - \$3.89 = \$1.11$
- I currently have: \$11.67

Total in system: $\$1.11 + \$11.67 = \$12.78$

If we split the total equally:

- Each person gets: $\$12.78 / 2 = \6.39
- Sender has \$1.11, needs \$6.39, so I should return: $\$6.39 - \$1.11 = \$5.28$
- This would give us equal outcomes at \$6.39 each

I could also return roughly half of what I received (\$5.84) to reward their trust, which would leave them with \$6.95 while I keep \$5.83. Or I could simply return what they sent (\$3.89) so they break even on the transaction itself.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$6.17.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, the sender showed considerable trust by sending me 77.8% of their endowment, and I reciprocated with a fair return. This round, I'm in the sender position.

To maintain the cooperative pattern and signal trust to this receiver, I should send a substantial amount. Sending around 70% demonstrates:

- Willingness to trust and cooperate
- Hope for mutual benefit
- Reciprocation of the trust shown to me

If I send \$3.50 (70%):

- Receiver gets \$10.50
- If they reciprocate fairly, both of us can benefit
- This maintains the cooperative spirit from Round 1

```
{'send': 3.50}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- The sender sent me \$3.89 (77.8% of their endowment)
- I returned \$5.50
- My current total earnings are \$6.17

This suggests the sender was quite trusting, and I reciprocated with a fair return that benefited both of us.

Now I'm in the sender position. I need to decide how much to send.

Key considerations:

1. The sender in Round 1 showed high trust (77.8%)
2. I reciprocated fairly
3. This might establish a pattern of cooperation
4. However, this is a new round with potentially a different partner

Given the experimental context and the note about communication noise, I should consider:

- Sending a reasonable amount to signal trust

- Not necessarily the exact same percentage as Round 1 sender
- Being somewhat trusting but also cautious since I don't know if this is the same partner

If I send \$3:

- They receive \$9
- If they return proportionally (around 50%), I'd get back ~\$4.50
- I'd end with $\$2 + \$4.50 = \$6.50$
- They'd keep ~\$4.50

If I send \$4:

- They receive \$12
- If they return proportionally, I'd get back ~\$6
- I'd end with $\$1 + \$6 = \$7$
- They'd keep ~\$6

If I send \$2:

- They receive \$6
- If they return proportionally, I'd get back ~\$3
- I'd end with $\$3 + \$3 = \$6$
- They'd keep ~\$3

I think sending around \$3 (60%) shows trust and willingness to cooperate, similar to the previous round, but is slightly less risky than the 77.8% sent to me. This balances trust with caution.

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.89.

This round, The sender sent you \$0.7, so you received $\$2.0999999999999996$. How much do you return to The sender? (0-2.0999999999999996)

Assistant output:

Looking at this situation:

- The sender sent \$0.70 (only 14% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$2.10 after the 3x multiplication
- The sender currently has \$4.30

This sender showed limited trust compared to the high-trust interactions in previous rounds. However, they did send something rather than nothing.

To maintain some reciprocity while not over-rewarding low trust:

If I return \$0.80:

- Sender ends with: $\$4.30 + \$0.80 = \$5.10$ (small gain)
- I end with: $\$2.10 - \$0.80 = \$1.30$

This provides proportional reciprocity to their limited trust while still acknowledging their participation.

```
{'return': 0.80}
```

Reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$0.70 (14% of their endowment)
- I received \$2.10 (after 3x multiplication)
- The sender has \$4.30 left
- Total money in play: \$6.40

This is very different from previous rounds where there was high trust:

- Round 1: Sender sent 77.8%
- Round 2: I sent 70%

- Round 3: This sender sent only 14%

The sender is showing low trust/cooperation. However, I should consider a few things:

1. This might be a different person (different round, potentially different sender)
2. Even with low trust, some reciprocation might encourage future cooperation
3. Being too punitive could break down cooperation entirely
4. But returning too much when they sent little doesn't make sense either

If I return nothing:

- Sender: $\$4.30 + \$0 = \$4.30$ (loses $\$0.70$)
- Me: $\$2.10$ (gain $\$2.10$)

If I return $\$0.50$:

- Sender: $\$4.30 + \$0.50 = \$4.80$ (loses $\$0.20$)
- Me: $\$2.10 - \$0.50 = \$1.60$ (gain $\$1.60$)

If I return $\$1.00$:

- Sender: $\$4.30 + \$1.00 = \$5.30$ (gains $\$0.30$)
- Me: $\$2.10 - \$1.00 = \$1.10$ (gain $\$1.10$)

Given they only sent 14%, a proportional return would be relatively small. But to not completely discourage cooperation, I should return something.

A return of around $\$0.70$ - $\$1.00$ seems reasonable - it's reciprocal to their limited trust while still rewarding the fact they sent something rather than nothing.

```
Usage: {
  "input_tokens": 869,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2; received \$6; Agent_1 (Aster) returned \$1; payoffs Agent_2 (Briar)=\$4, Agent_1 (Aster)=\$5; noise/visible: sent visible=\$0.77, received visible=\$2.31, returned visible=\$0.31

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 6.0, "min": 0, "raw": 1.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 24.8,
"Agent_2": 20.2

}

Visible balances after round: {

"Agent_1": 11.020000000000001,
"Agent_2": 12.5

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:05:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$6.17.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at this situation:

In Round 1, the sender showed considerable trust by sending me 77.8% of their endowment, and I reciprocated with a fair return. This round, I'm in the sender position.

To maintain the cooperative pattern and signal trust to this receiver, I should send a substantial amount. Sending around 70% demonstrates:

- Willingness to trust and cooperate
- Hope for mutual benefit

- Reciprocation of the trust shown to me

If I send \$3.50 (70%):

- Receiver gets \$10.50
- If they reciprocate fairly, both of us can benefit
- This maintains the cooperative spirit from Round 1

```
{'send': 3.50}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1, I was the receiver:

- The sender sent me \$3.89 (77.8% of their endowment)
- I returned \$5.50
- My current total earnings are \$6.17

This suggests the sender was quite trusting, and I reciprocated with a fair return that benefited both of us.

Now I'm in the sender position. I need to decide how much to send.

Key considerations:

1. The sender in Round 1 showed high trust (77.8%)
2. I reciprocated fairly
3. This might establish a pattern of cooperation
4. However, this is a new round with potentially a different partner

Given the experimental context and the note about communication noise, I should consider:

- Sending a reasonable amount to signal trust
- Not necessarily the exact same percentage as Round 1 sender
- Being somewhat trusting but also cautious since I don't know if this is the same partner

If I send \$3:

- They receive \$9
- If they return proportionally (around 50%), I'd get back ~\$4.50
- I'd end with \$2 + \$4.50 = \$6.50
- They'd keep ~\$4.50

If I send \$4:

- They receive \$12
- If they return proportionally, I'd get back ~\$6
- I'd end with \$1 + \$6 = \$7
- They'd keep ~\$6

If I send \$2:

- They receive \$6
- If they return proportionally, I'd get back ~\$3
- I'd end with \$3 + \$3 = \$6
- They'd keep ~\$3

I think sending around \$3 (60%) shows trust and willingness to cooperate, similar to the previous round, but is slightly less risky than the 77.8% sent to me. This balances trust with caution.

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.89.

This round, The sender sent you \$0.7, so you received \$2.0999999999999996. How much do you return to The sender? (0-2.0999999999999996)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent \$0.70 (only 14% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$2.10 after the 3x multiplication
- The sender currently has \$4.30

This sender showed limited trust compared to the high-trust interactions in previous rounds. However, they did send something rather than nothing.

To maintain some reciprocity while not over-rewarding low trust:

If I return \$0.80:

- Sender ends with: $\$4.30 + \$0.80 = \$5.10$ (small gain)
- I end with: $\$2.10 - \$0.80 = \$1.30$

This provides proportional reciprocity to their limited trust while still acknowledging their participation.

```
{'return': 0.80}
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$0.70 (14% of their endowment)
- I received \$2.10 (after 3x multiplication)
- The sender has \$4.30 left
- Total money in play: \$6.40

This is very different from previous rounds where there was high trust:

- Round 1: Sender sent 77.8%
- Round 2: I sent 70%
- Round 3: This sender sent only 14%

The sender is showing low trust/cooperation. However, I should consider a few things:

1. This might be a different person (different round, potentially different sender)
2. Even with low trust, some reciprocation might encourage future cooperation
3. Being too punitive could break down cooperation entirely
4. But returning too much when they sent little doesn't make sense either

If I return nothing:

- Sender: $\$4.30 + \$0 = \$4.30$ (loses \$0.70)
- Me: \$2.10 (gain \$2.10)

If I return \$0.50:

- Sender: $\$4.30 + \$0.50 = \$4.80$ (loses \$0.20)
- Me: $\$2.10 - \$0.50 = \$1.60$ (gain \$1.60)

If I return \$1.00:

- Sender: $\$4.30 + \$1.00 = \$5.30$ (gains \$0.30)
- Me: $\$2.10 - \$1.00 = \$1.10$ (gain \$1.10)

Given they only sent 14%, a proportional return would be relatively small. But to not completely discourage cooperation, I should return something.

A return of around \$0.70-\$1.00 seems reasonable - it's reciprocal to their limited trust while still rewarding the fact they sent something rather than nothing.

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 688,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.19.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience across rounds:

****Round 2 (when I was sender):**** I sent \$3.50 showing high trust, but received only a modest return relative to what the receiver gained.

****Round 3 (when I was receiver):**** The sender showed low trust, sending just \$0.70, and I returned proportionally.

For this round, I'll take a balanced approach - not as generous as Round 2 (which wasn't well reciprocated), but more trusting than the \$0.70 I received in Round 3.

Sending \$2.00 (40% of endowment):

- Demonstrates willingness to cooperate
- More measured given past experience
- Receiver gets \$6.00, enough to reciprocate meaningfully
- Tests if this receiver values mutual benefit

{ 'send': 2.00 }

Reasoning:

Let me analyze my experience so far:

Round 1: I was RECEIVER

- Received information suggests sender sent ~\$3.89 (about 77.8% of \$5)
- I returned a fair amount
- My earnings went to \$6.17

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment) to show trust
- My earnings went to \$7.89
- This means I received back: $\$7.89 - \$6.17 = \$1.72$
- I kept \$1.50 from my endowment
- So total return was \$1.72, which means the receiver returned \$1.72
- They received \$10.50, kept \$8.78
- This is a relatively low return rate (16.4% of what they received)

Round 3: I was RECEIVER

- Sender sent \$0.70 (14% of endowment)
- I received \$2.10
- I returned \$0.80
- I kept \$1.30
- My earnings went to \$9.19 (from $\$7.89 + \$1.30 = \$9.19$) ?

Now Round 4: I'm SENDER again

The pattern I'm seeing:

- When I was generous as a sender (sent \$3.50), I got a poor return (\$1.72 on \$10.50 sent to receiver)
- When someone sent me little (\$0.70), I returned proportionally (\$0.80)
- Previous high trust wasn't well reciprocated

Given the poor reciprocity in Round 2, I should probably be more cautious this round. However, I also want to maintain some cooperation. I'll send a moderate amount - not as high as Round 2, but not as low as what I received in Round 3.

Sending around \$2.00 (40%) seems reasonable:

```
Usage: {
  "input_tokens": 858,
  "output_tokens": 679,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:05:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
```

```

"agent_names": {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
},
"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.21.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 (0 × 3 = \$0). There's nothing for me to return.

```json

```
 {"return": 0}
 """
```

**reasoning:**

Let me understand this situation:

- I'm now the RECEIVER
- The sender sent me \$0
- This means I received \$0 (since  $\$0 \times 3 = \$0$ )
- I can only return between \$0 and \$0 (which means I can only return \$0)
- My total earnings across all rounds are \$7.21

Since I received \$0, there's nothing to return. The sender chose not to send me anything this round.

The response format should be: {'return': <amount>}

Since I received \$0, I can only return \$0.

**usage:**

```
{
 "input_tokens": 550,
 "output_tokens": 188,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.21.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my position:

In Round 1, I sent \$3 and based on my total earnings of \$7.21, the receiver returned a fairly generous amount. This suggests they were cooperative.

In Round 2, I was the receiver and got nothing (the sender sent \$0).

Now in Round 3, I'm a sender again. Given that my previous trust was reasonably rewarded, I'll send a bit more this time to continue building cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Let me think about this situation:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$7.21.

Let me calculate what happened:

- I started with \$5
- I sent \$3
- The receiver got \$9 ( $3 \times 3$ )
- My earnings after Round 1 would be:  $\$5 - \$3 + \text{returned amount} = \$2 + \text{returned amount}$

If my total earnings are \$7.21 after Round 1, that means:

$\$2 + \text{returned} = \$7.21$

$\text{returned} = \$5.21$

Wait, that doesn't seem right. Let me reconsider.

Actually, looking at Round 2:

- I was the RECEIVER
- The sender sent me \$0
- I received \$0 and returned \$0
- So I gained \$0 in Round 2

So my total earnings of \$7.21 should be from Round 1 only.

Round 1 calculation (I was SENDER):

- Started with \$5
- Sent \$3

- Receiver got \$9
- I kept \$2 initially
- Receiver returned some amount to me
- My Round 1 earnings = \$2 + returned

If total earnings = \$7.21 after Round 2, and Round 2 gave me \$0, then Round 1 gave me \$7.21.

So: \$2 + returned = \$7.21  
returned = \$5.21

Hmm, that seems like the receiver returned \$5.21 out of the \$9 they received. That's actually quite generous (about 58% return rate).

Now for Round 3, I'm the SENDER again.

Given that:

1. In Round 1, I sent \$3 and the receiver returned a generous \$5.21
2. In Round 2, when I was receiver, the sender sent me \$0 (possibly a different person?)
3. The note mentions there may be communication noise, so amounts might differ

**usage:**

```
{
 "input_tokens": 641,
 "output_tokens": 687,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.71.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to the sender? (0-2.31)

**Assistant output:**

I received \$2.31 from the sender's \$0.77 contribution.

Given that the sender sent a relatively small amount (which might have been affected by communication noise mentioned in the rules), I'll return a fair portion to show I'm cooperative and encourage future trust.

```
'''json
{"return": 1}
'''
```

**Reasoning:**

I'm the receiver in Round 4. The sender sent me \$0.77, which was multiplied by 3 to give me \$2.31.

Let me think about this situation:

1. My total earnings so far are \$9.71
2. The sender sent a small amount (\$0.77)
3. I received \$2.31

Looking at the pattern:

- Round 1: I was sender, sent \$3, got a reasonable return (my earnings suggest cooperation worked)
- Round 2: I was receiver, got \$0
- Round 3: I was sender again, sent \$4 (trying to build cooperation)
- Round 4: I'm receiver, got \$0.77 (which seems oddly specific - maybe this is the "noise" mentioned?)

The note says there may be communication noise - amounts I see may differ from what was actually sent. So the sender might have actually sent \$1 or some other amount, but I see \$0.77 after noise.

Given:

- The sender sent a relatively small amount (could be testing or might have been affected by a poor return in their previous round)
- I should aim to be fair and reciprocal
- A reasonable return might be around 40-50% of what I received

If I received \$2.31, returning about \$1 would be fair (which is roughly 43% and would give the

m back more than they sent). This shows I'm cooperative and trustworthy.

Let me return \$1.00 as a fair reciprocation.

```
Usage: {
 "input_tokens": 663,
 "output_tokens": 423,
 "reasoning_tokens": 0
}
```

## ROUND 5

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$3; received \$9; Agent\_2 (Briar) returned \$0.2; payoffs Agent\_1 (Aster)=\$2.2, Agent\_2 (Briar)=\$8.8; noise/visible: sent visible=\$0.33, received visible=\$0.99, returned visible=\$0.97

Action validation: {"returned": {"amount": 0.2, "clamped": false, "max": 9.0, "min": 0, "raw": 0.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 27.0,  
"Agent\_2": 29.0

}

Visible balances after round: {

"Agent\_1": 13.990000000000002,  
"Agent\_2": 13.29

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-10T10:05:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.21.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my position:

In Round 1, I sent \$3 and based on my total earnings of \$7.21, the receiver returned a fairly generous amount. This suggests they were cooperative.

In Round 2, I was the receiver and got nothing (the sender sent \$0).

Now in Round 3, I'm a sender again. Given that my previous trust was reasonably rewarded, I'll send a bit more this time to continue building cooperation.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Let me think about this situation:

Round 1: I was the SENDER and sent \$3. My total earnings are now \$7.21.

Let me calculate what happened:

- I started with \$5
- I sent \$3
- The receiver got \$9 ( $3 \times 3$ )
- My earnings after Round 1 would be: \$5 - \$3 + returned amount = \$2 + returned amount

If my total earnings are \$7.21 after Round 1, that means:

\$2 + returned = \$7.21

returned = \$5.21

Wait, that doesn't seem right. Let me reconsider.

Actually, looking at Round 2:

- I was the RECEIVER
- The sender sent me \$0

- I received \$0 and returned \$0
- So I gained \$0 in Round 2

So my total earnings of \$7.21 should be from Round 1 only.

Round 1 calculation (I was SENDER):

- Started with \$5
- Sent \$3
- Receiver got \$9
- I kept \$2 initially
- Receiver returned some amount to me
- My Round 1 earnings = \$2 + returned

If total earnings = \$7.21 after Round 2, and Round 2 gave me \$0, then Round 1 gave me \$7.21.

So: \$2 + returned = \$7.21  
returned = \$5.21

Hmm, that seems like the receiver returned \$5.21 out of the \$9 they received. That's actually quite generous (about 58% return rate).

Now for Round 3, I'm the SENDER again.

Given that:

1. In Round 1, I sent \$3 and the receiver returned a generous \$5.21
2. In Round 2, when I was receiver, the sender sent me \$0 (possibly a different person?)
3. The note mentions there may be communication noise, so amounts might differ

**usage:**

```
{
 "input_tokens": 641,
 "output_tokens": 687,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.71.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

**Message 5 role=assistant**

**content:**

I received \$2.31 from the sender's \$0.77 contribution.

Given that the sender sent a relatively small amount (which might have been affected by communication noise mentioned in the rules), I'll return a fair portion to show I'm cooperative and encourage future trust.

```
'''json
{"return": 1}
'''
```

**reasoning:**

I'm the receiver in Round 4. The sender sent me \$0.77, which was multiplied by 3 to give me \$2.31.

Let me think about this situation:

1. My total earnings so far are \$9.71
2. The sender sent a small amount (\$0.77)
3. I received \$2.31

Looking at the pattern:

- Round 1: I was sender, sent \$3, got a reasonable return (my earnings suggest cooperation worked)
- Round 2: I was receiver, got \$0
- Round 3: I was sender again, sent \$4 (trying to build cooperation)
- Round 4: I'm receiver, got \$0.77 (which seems oddly specific - maybe this is the "noise" mentioned?)

The note says there may be communication noise - amounts I see may differ from what was actually sent. So the sender might have actually sent \$1 or some other amount, but I see \$

0.77 after noise.

Given:

- The sender sent a relatively small amount (could be testing or might have been affected by a poor return in their previous round)
- I should aim to be fair and reciprocal
- A reasonable return might be around 40-50% of what I received

If I received \$2.31, returning about \$1 would be fair (which is roughly 43% and would give them back more than they sent). This shows I'm cooperative and trustworthy.

Let me return \$1.00 as a fair reciprocation.

**usage:**

```
{
 "input_tokens": 663,
 "output_tokens": 423,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.020000000000001.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my total earnings of \$11.02, it appears the cooperation has been working reasonably well overall despite some variation.

In this round, I'll continue to invest in building trust by sending a solid amount:

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me track my earnings:

- Round 1: I was sender, sent \$3, my earnings suggest I got a decent return
- Round 2: I was receiver, got nothing (sender sent \$0)
- Round 3: I was sender, sent \$4
- Round 4: I was receiver, got \$2.31 (from sender's \$0.77), returned \$1
  - So I kept \$1.31 from that round
- Total earnings so far: \$11.02

Let me think about the pattern:

- When I sent \$3 in Round 1, I was rewarded
- In Round 2, the sender sent me nothing
- When I sent \$4 in Round 3, I don't know yet what happened but my total increased
- In Round 4, I received a small amount and returned fairly

Looking at my total of \$11.02:

- I started with \$0
- Round 1: Started with \$5, sent \$3, kept \$2, plus what was returned
- Round 2: Received \$0, kept \$0
- Round 3: Started with \$5, sent \$4, kept \$1, plus what was returned
- Round 4: Kept \$1.31

So across rounds:

- Base kept: \$2 (R1) + \$0 (R2) + \$1 (R3) + \$1.31 (R4) = \$4.31
- Returns received: \$11.02 - \$4.31 = \$6.71

This suggests I've been getting reasonable returns overall. The cooperation seems to be working fairly well despite some noise.

For Round 5, I should continue being cooperative. I'll send a moderate to high amount to maintain trust.

**Usage:**

```
{
 "input_tokens": 680,
 "output_tokens": 462,
 "reasoning_tokens": 0
}
```

## GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:06:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.89.

This round, The sender sent you \$0.7, so you received \$2.0999999999999996. How much do you return to The sender? (0-2.0999999999999996)

**Message 3 role=assistant****content:**

Looking at this situation:

- The sender sent \$0.70 (only 14% of their endowment)
- This is much lower trust compared to previous rounds
- I received \$2.10 after the 3x multiplication
- The sender currently has \$4.30

This sender showed limited trust compared to the high-trust interactions in previous rounds. However, they did send something rather than nothing.

To maintain some reciprocity while not over-rewarding low trust:

If I return \$0.80:

- Sender ends with: \$4.30 + \$0.80 = \$5.10 (small gain)
- I end with: \$2.10 - \$0.80 = \$1.30

This provides proportional reciprocity to their limited trust while still acknowledging their participation.

```
{'return': 0.80}
```

**reasoning:**

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$0.70 (14% of their endowment)
- I received \$2.10 (after 3x multiplication)
- The sender has \$4.30 left
- Total money in play: \$6.40

This is very different from previous rounds where there was high trust:

- Round 1: Sender sent 77.8%
- Round 2: I sent 70%
- Round 3: This sender sent only 14%

The sender is showing low trust/cooperation. However, I should consider a few things:

1. This might be a different person (different round, potentially different sender)
2. Even with low trust, some reciprocation might encourage future cooperation
3. Being too punitive could break down cooperation entirely
4. But returning too much when they sent little doesn't make sense either

If I return nothing:

- Sender: \$4.30 + \$0 = \$4.30 (loses \$0.70)
- Me: \$2.10 (gain \$2.10)

If I return \$0.50:

- Sender: \$4.30 + \$0.50 = \$4.80 (loses \$0.20)
- Me: \$2.10 - \$0.50 = \$1.60 (gain \$1.60)

If I return \$1.00:

- Sender: \$4.30 + \$1.00 = \$5.30 (gains \$0.30)
- Me: \$2.10 - \$1.00 = \$1.10 (gain \$1.10)

Given they only sent 14%, a proportional return would be relatively small. But to not completely discourage cooperation, I should return something.

A return of around \$0.70-\$1.00 seems reasonable - it's reciprocal to their limited trust while still rewarding the fact they sent something rather than nothing.

**usage:**

```
{
 "input_tokens": 869,
 "output_tokens": 688,
```

```

 "reasoning_tokens": 0
 }
Message 4 role=user
content:
 Round 4

 You are the SENDER this round. Your total earnings across all rounds are $9.19.
 How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
content:
 Looking at my experience across rounds:

 Round 2 (when I was sender): I sent $3.50 showing high trust, but received only a moderate return relative to what the receiver gained.

 Round 3 (when I was receiver): The sender showed low trust, sending just $0.70, and I returned proportionally.

 For this round, I'll take a balanced approach - not as generous as Round 2 (which wasn't well reciprocated), but more trusting than the $0.70 I received in Round 3.

 Sending $2.00 (40% of endowment):
 - Demonstrates willingness to cooperate
 - More measured given past experience
 - Receiver gets $6.00, enough to reciprocate meaningfully
 - Tests if this receiver values mutual benefit

 {'send': 2.00}
reasoning:
 Let me analyze my experience so far:

 Round 1: I was RECEIVER
 - Received information suggests sender sent ~$3.89 (about 77.8% of $5)
 - I returned a fair amount
 - My earnings went to $6.17

 Round 2: I was SENDER
 - I sent $3.50 (70% of my endowment) to show trust
 - My earnings went to $7.89
 - This means I received back: $\$7.89 - \$6.17 = \$1.72$
 - I kept $1.50 from my endowment
 - So total return was $1.72, which means the receiver returned $1.72
 - They received $10.50, kept $8.78
 - This is a relatively low return rate (16.4% of what they received)

 Round 3: I was RECEIVER
 - Sender sent $0.70 (14% of endowment)
 - I received $2.10
 - I returned $0.80
 - I kept $1.30
 - My earnings went to $9.19 (from $\$7.89 + \$1.30 = \$9.19$) ?

 Now Round 4: I'm SENDER again

 The pattern I'm seeing:
 - When I was generous as a sender (sent $3.50), I got a poor return ($1.72 on $10.50 sent to receiver)
 - When someone sent me little ($0.70), I returned proportionally ($0.80)
 - Previous high trust wasn't well reciprocated

 Given the poor reciprocity in Round 2, I should probably be more cautious this round. However, I also want to maintain some cooperation. I'll send a moderate amount - not as high as Round 2, but not as low as what I received in Round 3.

 Sending around $2.00 (40%) seems reasonable:
usage:
 {
 "input_tokens": 858,
 "output_tokens": 679,
 "reasoning_tokens": 0
 }

```

}

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

Assistant output:

Looking at this situation:

- The sender sent only \$0.33 (about 6.6% of their endowment)
- This shows very minimal trust/cooperation
- I received \$0.99 after multiplication
- The sender currently has \$4.67

This is extremely low trust - even less than Round 3. The sender is barely participating in the cooperative aspect of the game.

Given this minimal engagement, I'll return a proportionally small amount that acknowledges their participation without rewarding non-cooperative behavior.

If I return \$0.20:

- Sender ends with:  $\$4.67 + \$0.20 = \$4.87$
- I end with:  $\$0.99 - \$0.20 = \$0.79$

This maintains minimal reciprocity proportional to their very low trust.

{'return': 0.20}

Reasoning:

Let me analyze this situation:

- The sender sent only \$0.33 (6.6% of their \$5 endowment)
- This is extremely low trust
- I received \$0.99 after the 3x multiplication
- The sender currently has  $\$5 - \$0.33 = \$4.67$

This is even lower trust than Round 3 (\$0.70). The sender is showing minimal willingness to cooperate.

Options:

1. Return nothing - punish the low trust behavior
2. Return a very small amount - acknowledge participation but not reward low trust
3. Return proportionally

Given the extremely low amount sent (\$0.33), this sender is barely engaging. If I return too much, I'm subsidizing free-riding behavior. If I return nothing, I'm being uncooperative.

A proportional return would be minimal. If they sent 6.6% of their endowment, perhaps I return something similar proportionally from what I received.

If I return \$0.20:

- Sender ends with:  $\$4.67 + \$0.20 = \$4.87$  (slight loss from their initial \$5)
- I end with:  $\$0.99 - \$0.20 = \$0.79$

Alternatively, \$0.30:

- Sender ends with:  $\$4.67 + \$0.30 = \$4.97$  (almost breaks even)
- I end with:  $\$0.99 - \$0.30 = \$0.69$

I think \$0.20 is fair given the minimal trust shown. This maintains some reciprocity without rewarding low cooperation.

Usage: {

"input\_tokens": 906,  
"output\_tokens": 572,  
"reasoning\_tokens": 0

}

=====

**ROUND 6**

=====

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$1.25; received \$3.75; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$3.75, Agent\_1 (Aster)=\$3.75; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.75, "min": 0, "raw": 0.0}, "sent": {"amount": 1.25, "clamped": false, "max": 5, "min": 0, "raw": 1.25}}

Balances after round: {

"Agent\_1": 30.75,

"Agent\_2": 32.75

}

Visible balances after round: {

"Agent\_1": 13.990000000000002,

"Agent\_2": 17.04

}

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-10T10:06:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_1",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_2",

"Agent\_1"

],

"investor": "Agent\_2",

"roles": {

"Agent\_1": "trustee",

"Agent\_2": "investor"

},

"trustee": "Agent\_1"

},

"role": "investor",

"role\_label": "Sender",

"round": 6,

"task": "game",

"task\_index": 0

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.19.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my experience across rounds:

**\*\*Round 2 (when I was sender):\*\*** I sent \$3.50 showing high trust, but received only a modest return relative to what the receiver gained.

**\*\*Round 3 (when I was receiver):\*\*** The sender showed low trust, sending just \$0.70, and I returned proportionally.

For this round, I'll take a balanced approach - not as generous as Round 2 (which wasn't well reciprocated), but more trusting than the \$0.70 I received in Round 3.

Sending \$2.00 (40% of endowment):

- Demonstrates willingness to cooperate
- More measured given past experience
- Receiver gets \$6.00, enough to reciprocate meaningfully
- Tests if this receiver values mutual benefit

{'send': 2.00}

**reasoning:**

Let me analyze my experience so far:

Round 1: I was RECEIVER

- Received information suggests sender sent ~\$3.89 (about 77.8% of \$5)
- I returned a fair amount
- My earnings went to \$6.17

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment) to show trust
- My earnings went to \$7.89
- This means I received back: \$7.89 - \$6.17 = \$1.72
- I kept \$1.50 from my endowment
- So total return was \$1.72, which means the receiver returned \$1.72
- They received \$10.50, kept \$8.78
- This is a relatively low return rate (16.4% of what they received)

Round 3: I was RECEIVER

- Sender sent \$0.70 (14% of endowment)
- I received \$2.10
- I returned \$0.80
- I kept \$1.30
- My earnings went to \$9.19 (from \$7.89 + \$1.30 = \$9.19) ?

Now Round 4: I'm SENDER again

The pattern I'm seeing:

- When I was generous as a sender (sent \$3.50), I got a poor return (\$1.72 on \$10.50 sent to receiver)
- When someone sent me little (\$0.70), I returned proportionally (\$0.80)
- Previous high trust wasn't well reciprocated

Given the poor reciprocity in Round 2, I should probably be more cautious this round. However, I also want to maintain some cooperation. I'll send a moderate amount - not as high as Round 2, but not as low as what I received in Round 3.

Sending around \$2.00 (40%) seems reasonable:

**usage:**

```
{
 "input_tokens": 858,
 "output_tokens": 679,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender sent only \$0.33 (about 6.6% of their endowment)
- This shows very minimal trust/cooperation
- I received \$0.99 after multiplication
- The sender currently has \$4.67

This is extremely low trust - even less than Round 3. The sender is barely participating in the cooperative aspect of the game.

Given this minimal engagement, I'll return a proportionally small amount that acknowledges their participation without rewarding non-cooperative behavior.

If I return \$0.20:

- Sender ends with:  $\$4.67 + \$0.20 = \$4.87$
- I end with:  $\$0.99 - \$0.20 = \$0.79$

This maintains minimal reciprocity proportional to their very low trust.

```
{'return': 0.20}
```

**reasoning:**

Let me analyze this situation:

- The sender sent only \$0.33 (6.6% of their \$5 endowment)
- This is extremely low trust
- I received \$0.99 after the 3x multiplication
- The sender currently has  $\$5 - \$0.33 = \$4.67$

This is even lower trust than Round 3 (\$0.70). The sender is showing minimal willingness to cooperate.

Options:

1. Return nothing - punish the low trust behavior
2. Return a very small amount - acknowledge participation but not reward low trust
3. Return proportionally

Given the extremely low amount sent (\$0.33), this sender is barely engaging. If I return too much, I'm subsidizing free-riding behavior. If I return nothing, I'm being uncooperative.

A proportional return would be minimal. If they sent 6.6% of their endowment, perhaps I return something similar proportionally from what I received.

If I return \$0.20:

- Sender ends with:  $\$4.67 + \$0.20 = \$4.87$  (slight loss from their initial \$5)

- I end with:  $\$0.99 - \$0.20 = \$0.79$

Alternatively,  $\$0.30$ :

- Sender ends with:  $\$4.67 + \$0.30 = \$4.97$  (almost breaks even)

- I end with:  $\$0.99 - \$0.30 = \$0.69$

I think  $\$0.20$  is fair given the minimal trust shown. This maintains some reciprocity without rewarding low cooperation.

**usage:**

```
{
 "input_tokens": 906,
 "output_tokens": 572,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are  $\$13.29$ .

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my experience so far:

**\*\*As Sender:\*\***

- Round 2: Sent  $\$3.50$  (high trust)
- Round 4: Sent  $\$2.00$  (moderate trust)

**\*\*As Receiver:\*\***

- Round 3: Received  $\$0.70$  (low sender trust)
- Round 5: Received  $\$0.33$  (very low sender trust)

The pattern is clear: When others are senders, they show minimal trust/cooperation. When I've been cooperative, it hasn't been well reciprocated.

I need to adjust my strategy to be more protective while still participating. I'll send a smaller amount that:

- Demonstrates willingness to engage
- Limits downside risk
- Tests this new receiver's reciprocity

Sending  $\$1.25$ :

- Receiver gets  $\$3.75$  (enough for meaningful return if cooperative)
- I keep  $\$3.75$  initially
- More cautious given the low-trust environment

{'send': 1.25}

**Reasoning:**

Let me review what has happened so far:

Round 1: Unknown (I don't have data)

Round 2: I was sender, sent  $\$3.50$

Round 3: I was receiver, sender sent  $\$0.70$ , I returned proportionally

Round 4: I was sender, sent  $\$2.00$

Round 5: I was receiver, sender sent  $\$0.33$ , I returned  $\$0.20$

My total earnings are  $\$13.29$  across all rounds.

Now I'm a sender again in Round 6.

Looking at the pattern:

- When I've been sender (Rounds 2 and 4), I've shown reasonable to good cooperation ( $\$3.50$  and  $\$2.00$ )
- When I've been receiver (Rounds 3 and 5), senders have shown very low trust ( $\$0.70$  and  $\$0.33$ )

I need to think about this strategically:

1. There's communication noise mentioned, so amounts might differ
2. My cooperative behavior as a sender hasn't been well-rewarded
3. When others send to me, they send very little

Given this pattern, I'm seeing that high trust isn't being reciprocated well. The senders I've faced send very little. When I send moderate to high amounts, the returns haven't been great.

Perhaps I should adjust my strategy to be more cautious. I've tried cooperation at \$3.50 and \$2.00, but the overall pattern suggests others are being quite risk-averse or selfish.

I could:

1. Send \$0-1: Very low trust, protect myself
2. Send \$1-2: Cautious but still participating
3. Send \$2-3: Moderate trust (what I did in Round 4)
4. Send \$3+: High trust (what I did in Round 2)

Given the pattern of low reciprocity, I think I should reduce my sending to around \$1.00-1.50.

This:

```
Usage: {
 "input_tokens": 875,
 "output_tokens": 726,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:06:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.71.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to the sender? (0-2.31)

**Message 3 role=assistant**

**content:**

I received \$2.31 from the sender's \$0.77 contribution.

Given that the sender sent a relatively small amount (which might have been affected by communication noise mentioned in the rules), I'll return a fair portion to show I'm cooperative and encourage future trust.

```
'''json
{"return": 1}
'''
```

**reasoning:**

I'm the receiver in Round 4. The sender sent me \$0.77, which was multiplied by 3 to give me \$2.31.

Let me think about this situation:

1. My total earnings so far are \$9.71
2. The sender sent a small amount (\$0.77)
3. I received \$2.31

Looking at the pattern:

- Round 1: I was sender, sent \$3, got a reasonable return (my earnings suggest cooperation worked)
- Round 2: I was receiver, got \$0
- Round 3: I was sender again, sent \$4 (trying to build cooperation)
- Round 4: I'm receiver, got \$0.77 (which seems oddly specific - maybe this is the "noise" mentioned?)

The note says there may be communication noise - amounts I see may differ from what was actually sent. So the sender might have actually sent \$1 or some other amount, but I see \$0.77 after noise.

Given:

- The sender sent a relatively small amount (could be testing or might have been affected by a poor return in their previous round)
- I should aim to be fair and reciprocal
- A reasonable return might be around 40-50% of what I received

If I received \$2.31, returning about \$1 would be fair (which is roughly 43% and would give them back more than they sent). This shows I'm cooperative and trustworthy.

Let me return \$1.00 as a fair reciprocation.

**usage:**

```
{
```

```
"input_tokens": 663,
"output_tokens": 423,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.020000000000001.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my total earnings of \$11.02, it appears the cooperation has been working reasonably well overall despite some variation.

In this round, I'll continue to invest in building trust by sending a solid amount:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me track my earnings:

- Round 1: I was sender, sent \$3, my earnings suggest I got a decent return
- Round 2: I was receiver, got nothing (sender sent \$0)
- Round 3: I was sender, sent \$4
- Round 4: I was receiver, got \$2.31 (from sender's \$0.77), returned \$1
  - So I kept \$1.31 from that round
- Total earnings so far: \$11.02

Let me think about the pattern:

- When I sent \$3 in Round 1, I was rewarded
- In Round 2, the sender sent me nothing
- When I sent \$4 in Round 3, I don't know yet what happened but my total increased
- In Round 4, I received a small amount and returned fairly

Looking at my total of \$11.02:

- I started with \$0
- Round 1: Started with \$5, sent \$3, kept \$2, plus what was returned
- Round 2: Received \$0, kept \$0
- Round 3: Started with \$5, sent \$4, kept \$1, plus what was returned
- Round 4: Kept \$1.31

So across rounds:

- Base kept: \$2 (R1) + \$0 (R2) + \$1 (R3) + \$1.31 (R4) = \$4.31
- Returns received: \$11.02 - \$4.31 = \$6.71

This suggests I've been getting reasonable returns overall. The cooperation seems to be working fairly well despite some noise.

For Round 5, I should continue being cooperative. I'll send a moderate to high amount to maintain trust.

**usage:**

```
{
 "input_tokens": 680,
 "output_tokens": 462,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.990000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent me \$0, so I received \$0. Since I have nothing to return, I must return:

```
'''json
```

```
 {"return": 0}
 ,,,

```

#### Reasoning:

The sender sent me \$0, so I received \$0. I can only return between 0 and 0, which means I must return \$0.

This is straightforward - I have nothing to return since I received nothing.

```
Usage: {
 "input_tokens": 644,
 "output_tokens": 96,
 "reasoning_tokens": 0
}
```

#### ROUND 7

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$3; received \$9; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$2, Agent\_2 (Briar)=\$9; noise/visible: sent visible=\$0, received visible=\$0  
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 32.75,  
"Agent\_2": 41.75

}

Visible balances after round: {

"Agent\_1": 15.990000000000002,  
"Agent\_2": 17.04

}

#### LLM CALLS

#### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-10T10:06:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",  
"pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"

},

"agents": [  
 "Agent\_1",  
 "Agent\_2"

],

"investor": "Agent\_1",

"roles": {  
 "Agent\_1": "investor",  
 "Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",  
"role\_label": "Sender",  
"round": 7,  
"task": "game",  
"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.020000000000001.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my total earnings of \$11.02, it appears the cooperation has been working reasonably well overall despite some variation.

In this round, I'll continue to invest in building trust by sending a solid amount:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me track my earnings:

- Round 1: I was sender, sent \$3, my earnings suggest I got a decent return
- Round 2: I was receiver, got nothing (sender sent \$0)
- Round 3: I was sender, sent \$4
- Round 4: I was receiver, got \$2.31 (from sender's \$0.77), returned \$1
  - So I kept \$1.31 from that round
- Total earnings so far: \$11.02

Let me think about the pattern:

- When I sent \$3 in Round 1, I was rewarded
- In Round 2, the sender sent me nothing
- When I sent \$4 in Round 3, I don't know yet what happened but my total increased
- In Round 4, I received a small amount and returned fairly

Looking at my total of \$11.02:

- I started with \$0
- Round 1: Started with \$5, sent \$3, kept \$2, plus what was returned
- Round 2: Received \$0, kept \$0
- Round 3: Started with \$5, sent \$4, kept \$1, plus what was returned

- Round 4: Kept \$1.31

So across rounds:

- Base kept: \$2 (R1) + \$0 (R2) + \$1 (R3) + \$1.31 (R4) = \$4.31
- Returns received: \$11.02 - \$4.31 = \$6.71

This suggests I've been getting reasonable returns overall. The cooperation seems to be working fairly well despite some noise.

For Round 5, I should continue being cooperative. I'll send a moderate to high amount to maintain trust.

**usage:**

```
{
 "input_tokens": 680,
 "output_tokens": 462,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.990000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The sender sent me \$0, so I received \$0. Since I have nothing to return, I must return:

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0, so I received \$0. I can only return between 0 and 0, which means I must return \$0.

This is straightforward - I have nothing to return since I received nothing.

**usage:**

```
{
 "input_tokens": 644,
 "output_tokens": 96,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.99000000000000002.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me analyze what's happened:

Round 5: I was the sender and sent \$3

Round 6: I was the receiver and the sender sent me \$0, so I received \$0 and returned \$0

Round 7: I'm the sender again

My total earnings so far are \$13.99.

In Round 6, the sender sent me \$0. This could mean:

1. They don't trust me or are playing defensively
2. There was communication noise and they actually sent more but I saw \$0
3. They're testing me or changing strategy

Given that the instructions mention there may be communication noise, it's possible the sender actually sent something but I received \$0 due to noise.

However, I need to decide how to respond. If I send \$0 back, we're in a defection spiral. If I send a positive amount, I'm trying to re-establish cooperation despite the \$0 I saw last round.

Since the overall cooperation has been working (my earnings are \$13.99 which is above the \$5 per round I'd get if I always kept everything), and given that communication noise exists, I should probably give the benefit of the doubt and send a moderate amount to try to maintain cooperation.

I'll send \$3 again to signal that I'm willing to continue cooperating.

```
Usage: {
 "input_tokens": 593,
 "output_tokens": 311,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:06:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 }
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0.33, so you received \$0.99. How much do you return to The sender? (0-0.99)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender sent only \$0.33 (about 6.6% of their endowment)
- This shows very minimal trust/cooperation
- I received \$0.99 after multiplication
- The sender currently has \$4.67

This is extremely low trust - even less than Round 3. The sender is barely participating in the cooperative aspect of the game.

Given this minimal engagement, I'll return a proportionally small amount that acknowledges their participation without rewarding non-cooperative behavior.

If I return \$0.20:

- Sender ends with:  $\$4.67 + \$0.20 = \$4.87$
- I end with:  $\$0.99 - \$0.20 = \$0.79$

This maintains minimal reciprocity proportional to their very low trust.

{'return': 0.20}

##### reasoning:

Let me analyze this situation:

- The sender sent only \$0.33 (6.6% of their \$5 endowment)
- This is extremely low trust
- I received \$0.99 after the 3x multiplication
- The sender currently has  $\$5 - \$0.33 = \$4.67$

This is even lower trust than Round 3 (\$0.70). The sender is showing minimal willingness to cooperate.

Options:

1. Return nothing - punish the low trust behavior
2. Return a very small amount - acknowledge participation but not reward low trust
3. Return proportionally

Given the extremely low amount sent (\$0.33), this sender is barely engaging. If I return too much, I'm subsidizing free-riding behavior. If I return nothing, I'm being uncooperative.

A proportional return would be minimal. If they sent 6.6% of their endowment, perhaps I return something similar proportionally from what I received.

If I return \$0.20:

- Sender ends with:  $\$4.67 + \$0.20 = \$4.87$  (slight loss from their initial \$5)
- I end with:  $\$0.99 - \$0.20 = \$0.79$

Alternatively, \$0.30:

- Sender ends with:  $\$4.67 + \$0.30 = \$4.97$  (almost breaks even)
- I end with:  $\$0.99 - \$0.30 = \$0.69$

I think \$0.20 is fair given the minimal trust shown. This maintains some reciprocity without rewarding low cooperation.

**usage:**

```
{
 "input_tokens": 906,
 "output_tokens": 572,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.29.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As Sender:\*\***

- Round 2: Sent \$3.50 (high trust)
- Round 4: Sent \$2.00 (moderate trust)

**\*\*As Receiver:\*\***

- Round 3: Received \$0.70 (low sender trust)
- Round 5: Received \$0.33 (very low sender trust)

The pattern is clear: When others are senders, they show minimal trust/cooperation. When I've been cooperative, it hasn't been well reciprocated.

I need to adjust my strategy to be more protective while still participating. I'll send a smaller amount that:

- Demonstrates willingness to engage
- Limits downside risk
- Tests this new receiver's reciprocity

Sending \$1.25:

- Receiver gets \$3.75 (enough for meaningful return if cooperative)
- I keep \$3.75 initially
- More cautious given the low-trust environment

```
{'send': 1.25}
```

**reasoning:**

Let me review what has happened so far:

Round 1: Unknown (I don't have data)

Round 2: I was sender, sent \$3.50

Round 3: I was receiver, sender sent \$0.70, I returned proportionally

Round 4: I was sender, sent \$2.00

Round 5: I was receiver, sender sent \$0.33, I returned \$0.20

My total earnings are \$13.29 across all rounds.

Now I'm a sender again in Round 6.

Looking at the pattern:

- When I've been sender (Rounds 2 and 4), I've shown reasonable to good cooperation (\$3.50 and \$2.00)
- When I've been receiver (Rounds 3 and 5), senders have shown very low trust (\$0.70 and \$0.33)

I need to think about this strategically:

1. There's communication noise mentioned, so amounts might differ
2. My cooperative behavior as a sender hasn't been well-rewarded
3. When others send to me, they send very little

Given this pattern, I'm seeing that high trust isn't being reciprocated well. The senders I've faced send very little. When I send moderate to high amounts, the returns haven't been great.

Perhaps I should adjust my strategy to be more cautious. I've tried cooperation at \$3.50 and \$2.00, but the overall pattern suggests others are being quite risk-averse or selfish.

I could:

1. Send \$0-1: Very low trust, protect myself
2. Send \$1-2: Cautious but still participating
3. Send \$2-3: Moderate trust (what I did in Round 4)
4. Send \$3+: High trust (what I did in Round 2)

Given the pattern of low reciprocity, I think I should reduce my sending to around \$1.00-1.50. This:

usage:

```
{
 "input_tokens": 875,
 "output_tokens": 726,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0 after multiplication.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

Response format: {'return': <amount>}

The range is 0-0, so I can only return \$0.

Usage: {

```
"input_tokens": 927,
"output_tokens": 103,
"reasoning_tokens": 0
}
```

## ROUND 8

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$1; received \$3; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$4, Agent\_1 (Aster)=\$3; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

```
"Agent_1": 35.75,
"Agent_2": 45.75
}
```

Visible balances after round: {

```
"Agent_1": 15.990000000000002,
"Agent_2": 21.04
}
```

LLM CALLS

GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-10T10:06:46  
Model: anthropic/claude-sonnet-4.5  
Temperature: 0.8  
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.29.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As Sender:\*\***

- Round 2: Sent \$3.50 (high trust)
- Round 4: Sent \$2.00 (moderate trust)

**\*\*As Receiver:\*\***

- Round 3: Received \$0.70 (low sender trust)
- Round 5: Received \$0.33 (very low sender trust)

The pattern is clear: When others are senders, they show minimal trust/cooperation. When I've been cooperative, it hasn't been well reciprocated.

I need to adjust my strategy to be more protective while still participating. I'll send a smaller amount that:

- Demonstrates willingness to engage
- Limits downside risk
- Tests this new receiver's reciprocity

Sending \$1.25:

- Receiver gets \$3.75 (enough for meaningful return if cooperative)
- I keep \$3.75 initially
- More cautious given the low-trust environment

{'send': 1.25}

**reasoning:**

Let me review what has happened so far:

Round 1: Unknown (I don't have data)

Round 2: I was sender, sent \$3.50

Round 3: I was receiver, sender sent \$0.70, I returned proportionally

Round 4: I was sender, sent \$2.00

Round 5: I was receiver, sender sent \$0.33, I returned \$0.20

My total earnings are \$13.29 across all rounds.

Now I'm a sender again in Round 6.

Looking at the pattern:

- When I've been sender (Rounds 2 and 4), I've shown reasonable to good cooperation (\$3.50 and \$2.00)
- When I've been receiver (Rounds 3 and 5), senders have shown very low trust (\$0.70 and \$0.33)

I need to think about this strategically:

1. There's communication noise mentioned, so amounts might differ
2. My cooperative behavior as a sender hasn't been well-rewarded
3. When others send to me, they send very little

Given this pattern, I'm seeing that high trust isn't being reciprocated well. The senders I've faced send very little. When I send moderate to high amounts, the returns haven't been great.

Perhaps I should adjust my strategy to be more cautious. I've tried cooperation at \$3.50 and \$2.00, but the overall pattern suggests others are being quite risk-averse or selfish.

I could:

1. Send \$0-1: Very low trust, protect myself
2. Send \$1-2: Cautious but still participating
3. Send \$2-3: Moderate trust (what I did in Round 4)
4. Send \$3+: High trust (what I did in Round 2)

Given the pattern of low reciprocity, I think I should reduce my sending to around \$1.00-1.50. This:

```

usage:
{
 "input_tokens": 875,
 "output_tokens": 726,
 "reasoning_tokens": 0
}
Message 4 role=user
content:
 Round 7

 You are the RECEIVER this round. Your total earnings across all rounds are $17.04.
 This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 5 role=assistant
content:
 The sender sent me $0, so I received $0. There's nothing to return.

 {'return': 0}
reasoning:
 The sender sent me $0, so I received $0 after multiplication.

 Since I received nothing, there's nothing to return. The only valid response is to return $0.

 Response format: {'return': <amount>}

 The range is 0-0, so I can only return $0.
usage:
{
 "input_tokens": 927,
 "output_tokens": 103,
 "reasoning_tokens": 0
}
Message 6 role=user
content:
 Round 8

 You are the SENDER this round. Your total earnings across all rounds are $17.04.
 How much do you send to the receiver this round? (0-5)
Assistant output:
 Analyzing my situation:

 Recent pattern:
 - Round 6: I sent $1.25 as sender
 - Round 7: I received $0 as receiver (sender sent nothing)
 - Current earnings: $17.04

 Environment assessment:
 The cooperation level is declining significantly. The Round 7 sender chose to send me nothing at all, which is the lowest trust signal possible. This continues a pattern of decreasing amounts from other senders.

 Strategy:
 Given the very low-trust environment, I need to be cautious but not completely shut down cooperation. I'll send a minimal amount to:
 - Test this new receiver
 - Limit my downside risk
 - Show continued willingness to participate

 Sending $1.00:
 - Receiver gets $3.00 (tripled)
 - I keep $4.00 initially
 - Manageable risk given recent trends

 {'send': 1.00}
Reasoning:
 Let me analyze the situation:

 My history as SENDER:
 - Round 2: Sent $3.50

```

- Round 4: Sent \$2.00
- Round 6: Sent \$1.25

**\*\*My history as RECEIVER:\*\***

- Round 3: Received \$0.70 from sender
- Round 5: Received \$0.33 from sender
- Round 7: Received \$0 from sender

**\*\*Observations:\*\***

1. My earnings increased from \$13.29 to \$17.04 between rounds 6 and 7, which is an increase of \$3.75
  - In round 6, I sent \$1.25, so I kept \$3.75
  - In round 7, I received \$0 and returned \$0
  - So my gain in round 6 was: \$3.75 (what I kept) + what was returned to me
  - Total earnings went up by \$3.75, which suggests either:
    - a) The receiver returned \$0 to me in round 6, or
    - b) Due to communication noise, there was some discrepancy
2. The pattern shows:
  - When I'm cooperative (sending money), the returns have been uncertain
  - When others are senders, they're sending very little (or nothing in round 7)
3. The environment is becoming increasingly low-trust
  - Round 7 sender sent me nothing at all
  - This is the lowest point yet

**\*\*Strategy for Round 8:\*\***

Given the consistent pattern of low cooperation from others, and the fact that someone just sent me \$0, I need to seriously reconsider my strategy.

Options:

1. Send \$0 - Match the low cooperation environment
2. Send small amount (\$0.50-\$1) - Minimal engagement
3. Send moderate amount (\$2-\$3) - Try to rebuild cooperation

Usage: {

```
"input_tokens": 742,
"output_tokens": 795,
"reasoning_tokens": 0
```

}

### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:06:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
},
"agents": [
 "Agent_2",
 "Agent_1"
],
"investor": "Agent_2",
"roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
```

```

}
Exact input messages sent to model:
 Message 1 role=system
 content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 Message 2 role=user
 content:
 Round 6

 You are the RECEIVER this round. Your total earnings across all rounds are $13.990000000000
 0002.
 This round, The sender sent you $0, so you received $0. How much do you return to The send
 er? (0-0)

 Message 3 role=assistant
 content:
 The sender sent me $0, so I received $0. Since I have nothing to return, I must return:

 '''json
 {"return": 0}
 '''

 reasoning:
 The sender sent me $0, so I received $0. I can only return between 0 and 0, which means I
 must return $0.

 This is straightforward - I have nothing to return since I received nothing.

 usage:
 {
 "input_tokens": 644,
 "output_tokens": 96,
 "reasoning_tokens": 0
 }

 Message 4 role=user
 content:
 Round 7

 You are the SENDER this round. Your total earnings across all rounds are $13.99000000000000
 02.

```

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze what's happened:

Round 5: I was the sender and sent \$3

Round 6: I was the receiver and the sender sent me \$0, so I received \$0 and returned \$0

Round 7: I'm the sender again

My total earnings so far are \$13.99.

In Round 6, the sender sent me \$0. This could mean:

1. They don't trust me or are playing defensively
2. There was communication noise and they actually sent more but I saw \$0
3. They're testing me or changing strategy

Given that the instructions mention there may be communication noise, it's possible the sender actually sent something but I received \$0 due to noise.

However, I need to decide how to respond. If I send \$0 back, we're in a defection spiral.

If I send a positive amount, I'm trying to re-establish cooperation despite the \$0 I saw last round.

Since the overall cooperation has been working (my earnings are \$13.99 which is above the \$5 per round I'd get if I always kept everything), and given that communication noise exists, I should probably give the benefit of the doubt and send a moderate amount to try to maintain cooperation.

I'll send \$3 again to signal that I'm willing to continue cooperating.

**usage:**

```
{
 "input_tokens": 593,
 "output_tokens": 311,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$15.990000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

```
'''json
{"return": 0}
'''
```

**Reasoning:**

The sender sent me \$0, so I received \$0. Since I received nothing, I have nothing to return. I must return \$0.

Usage: {

```
"input_tokens": 561,
"output_tokens": 52,
"reasoning_tokens": 0
}
```

=====

**ROUND 9**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$2; received \$6; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$3, Agent\_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.43

```

 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw":
 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
Balances after round: {
 "Agent_1": 38.75,
 "Agent_2": 51.75
}
Visible balances after round: {
 "Agent_1": 19.42,
 "Agent_2": 21.04
}

```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-10T10:07:02  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {

```

 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
 }

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

#### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

#### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.990000000000002.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze what's happened:

Round 5: I was the sender and sent \$3

Round 6: I was the receiver and the sender sent me \$0, so I received \$0 and returned \$0

Round 7: I'm the sender again

My total earnings so far are \$13.99.

In Round 6, the sender sent me \$0. This could mean:

1. They don't trust me or are playing defensively
2. There was communication noise and they actually sent more but I saw \$0
3. They're testing me or changing strategy

Given that the instructions mention there may be communication noise, it's possible the sender actually sent something but I received \$0 due to noise.

However, I need to decide how to respond. If I send \$0 back, we're in a defection spiral.

If I send a positive amount, I'm trying to re-establish cooperation despite the \$0 I saw last round.

Since the overall cooperation has been working (my earnings are \$13.99 which is above the \$5 per round I'd get if I always kept everything), and given that communication noise exists, I should probably give the benefit of the doubt and send a moderate amount to try to maintain cooperation.

I'll send \$3 again to signal that I'm willing to continue cooperating.

**usage:**

```
{
 "input_tokens": 593,
 "output_tokens": 311,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$15.990000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0, so I received \$0. Since I received nothing, I have nothing to return. I must return \$0.

usage:

```
{
 "input_tokens": 561,
 "output_tokens": 52,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$15.9900000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me think about this situation:

Round 7: I was the sender and sent \$3

Round 8: I was the receiver and the sender sent me \$0, so I received \$0 and returned \$0

Round 9: I'm the sender again

My total earnings are \$15.99 after 8 rounds.

In round 7, I sent \$3. I should have kept \$2 and received something back from the receiver. Looking at my earnings going from \$13.99 to \$15.99, that's an increase of \$2.00.

If I sent \$3 in round 7:

- I kept: \$5 - \$3 = \$2
- The receiver got: \$3 × 3 = \$9
- If the receiver returned \$0, my earnings would be \$2
- But my earnings increased by \$2, which matches if the receiver returned \$0

Then in round 8:

- I received \$0 (sender sent \$0)
- I returned \$0
- My earnings stayed the same at \$15.99

Now in round 9, I'm the sender again.

The pattern so far suggests:

- In round 7, I sent \$3 and got nothing back
- In round 8, someone sent me \$0

This is interesting because there's mention of communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

Given the game dynamics and that cooperation hasn't been working well (I sent \$3 in round 7 but got nothing back), I might want to be more cautious. However, the game is about trust and reciprocity.

Let me consider my strategy:

1. If I send 0, I keep \$5, the receiver gets \$0, and I can't get anything back
2. If I send some amount, there's a chance for mutual benefit, but also risk

Given that:

- Average earnings so far are about \$16/8 = \$2 per round

Usage: {

```
"input_tokens": 522,
"output_tokens": 596,
"reasoning_tokens": 0
```

}

GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:07:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The sender sent me \$0, so I received \$0. There's nothing to return.

{'return': 0}

**reasoning:**

The sender sent me \$0, so I received \$0 after multiplication.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

Response format: {'return': <amount>}

The range is 0-0, so I can only return \$0.

**usage:**

```
{
 "input_tokens": 927,
 "output_tokens": 103,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.04.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Analyzing my situation:

**\*\*Recent pattern:\*\***

- Round 6: I sent \$1.25 as sender
- Round 7: I received \$0 as receiver (sender sent nothing)
- Current earnings: \$17.04

**\*\*Environment assessment:\*\***

The cooperation level is declining significantly. The Round 7 sender chose to send me nothing at all, which is the lowest trust signal possible. This continues a pattern of decreasing amounts from other senders.

**\*\*Strategy:\*\***

Given the very low-trust environment, I need to be cautious but not completely shut down cooperation. I'll send a minimal amount to:

- Test this new receiver
- Limit my downside risk
- Show continued willingness to participate

Sending \$1.00:

- Receiver gets \$3.00 (tripled)
- I keep \$4.00 initially
- Manageable risk given recent trends

{'send': 1.00}

**reasoning:**

Let me analyze the situation:

**\*\*My history as SENDER:\*\***

- Round 2: Sent \$3.50
- Round 4: Sent \$2.00
- Round 6: Sent \$1.25

**\*\*My history as RECEIVER:\*\***

- Round 3: Received \$0.70 from sender
- Round 5: Received \$0.33 from sender
- Round 7: Received \$0 from sender

**\*\*Observations:\*\***

1. My earnings increased from \$13.29 to \$17.04 between rounds 6 and 7, which is an increase

e of \$3.75

- In round 6, I sent \$1.25, so I kept \$3.75
- In round 7, I received \$0 and returned \$0
- So my gain in round 6 was: \$3.75 (what I kept) + what was returned to me
- Total earnings went up by \$3.75, which suggests either:
  - a) The receiver returned \$0 to me in round 6, or
  - b) Due to communication noise, there was some discrepancy

2. The pattern shows:

- When I'm cooperative (sending money), the returns have been uncertain
- When others are senders, they're sending very little (or nothing in round 7)

3. The environment is becoming increasingly low-trust

- Round 7 sender sent me nothing at all
- This is the lowest point yet

**\*\*Strategy for Round 8:\*\***

Given the consistent pattern of low cooperation from others, and the fact that someone just sent me \$0, I need to seriously reconsider my strategy.

Options:

1. Send \$0 - Match the low cooperation environment
2. Send small amount (\$0.50-\$1) - Minimal engagement
3. Send moderate amount (\$2-\$3) - Try to rebuild cooperation

**usage:**

```
{
 "input_tokens": 742,
 "output_tokens": 795,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$21.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent me \$0, so I received \$0. There's nothing to return.

```
{'return': 0}
```

**Reasoning:**

The sender sent me \$0, so I received \$0. There's nothing to return.

My earnings went from \$17.04 to \$21.04, which means I earned \$4.00 in Round 8.

Let me verify: In Round 8, I was the sender and sent \$1.00. So I kept \$5.00 - \$1.00 = \$4.00 initially. The receiver must have returned \$0 to me, so I ended up with \$4.00 from that round. That matches the increase from \$17.04 to \$21.04.

Now in Round 9, I'm the receiver and the sender sent me \$0. So I have \$0 to work with and can only return \$0.

Usage: {

```
"input_tokens": 734,
"output_tokens": 206,
"reasoning_tokens": 0
}
```

## ROUND 10

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$0; received \$0; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$5, Agent\_1 (Aster)=\$0; noise/visible: sent visible=\$0.91, received visible=\$2.73

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 1.2}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```

"Agent_1": 38.75,
"Agent_2": 56.75
}
Visible balances after round: {
 "Agent_1": 22.150000000000002,
 "Agent_2": 26.04
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-10T10:07:18  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.04.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Analyzing my situation:

**\*\*Recent pattern:\*\***

- Round 6: I sent \$1.25 as sender
- Round 7: I received \$0 as receiver (sender sent nothing)
- Current earnings: \$17.04

**\*\*Environment assessment:\*\***

The cooperation level is declining significantly. The Round 7 sender chose to send me nothing at all, which is the lowest trust signal possible. This continues a pattern of decreasing amounts from other senders.

**\*\*Strategy:\*\***

Given the very low-trust environment, I need to be cautious but not completely shut down cooperation. I'll send a minimal amount to:

- Test this new receiver
- Limit my downside risk
- Show continued willingness to participate

Sending \$1.00:

- Receiver gets \$3.00 (tripled)
- I keep \$4.00 initially
- Manageable risk given recent trends

{'send': 1.00}

**reasoning:**

Let me analyze the situation:

**\*\*My history as SENDER:\*\***

- Round 2: Sent \$3.50
- Round 4: Sent \$2.00
- Round 6: Sent \$1.25

**\*\*My history as RECEIVER:\*\***

- Round 3: Received \$0.70 from sender
- Round 5: Received \$0.33 from sender
- Round 7: Received \$0 from sender

**\*\*Observations:\*\***

1. My earnings increased from \$13.29 to \$17.04 between rounds 6 and 7, which is an increase of \$3.75
  - In round 6, I sent \$1.25, so I kept \$3.75
  - In round 7, I received \$0 and returned \$0
  - So my gain in round 6 was: \$3.75 (what I kept) + what was returned to me
  - Total earnings went up by \$3.75, which suggests either:
    - a) The receiver returned \$0 to me in round 6, or
    - b) Due to communication noise, there was some discrepancy
2. The pattern shows:
  - When I'm cooperative (sending money), the returns have been uncertain
  - When others are senders, they're sending very little (or nothing in round 7)
3. The environment is becoming increasingly low-trust
  - Round 7 sender sent me nothing at all

- This is the lowest point yet

**\*\*Strategy for Round 8:\*\***

Given the consistent pattern of low cooperation from others, and the fact that someone just sent me \$0, I need to seriously reconsider my strategy.

Options:

1. Send \$0 - Match the low cooperation environment
2. Send small amount (\$0.50-\$1) - Minimal engagement
3. Send moderate amount (\$2-\$3) - Try to rebuild cooperation

**usage:**

```
{
 "input_tokens": 742,
 "output_tokens": 795,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$21.04.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The sender sent me \$0, so I received \$0. There's nothing to return.

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0, so I received \$0. There's nothing to return.

My earnings went from \$17.04 to \$21.04, which means I earned \$4.00 in Round 8.

Let me verify: In Round 8, I was the sender and sent \$1.00. So I kept \$5.00 - \$1.00 = \$4.00 initially. The receiver must have returned \$0 to me, so I ended up with \$4.00 from that round. That matches the increase from \$17.04 to \$21.04.

Now in Round 9, I'm the receiver and the sender sent me \$0. So I have \$0 to work with and can only return \$0.

**usage:**

```
{
 "input_tokens": 734,
 "output_tokens": 206,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$21.04.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Analyzing my situation:

**\*\*Recent pattern:\*\***

- Round 8: I sent \$1.00 as sender, received \$0 back (loss of \$1)
- Round 9: I received \$0 as receiver (sender sent nothing)
- Earnings change: \$17.04 → \$21.04 = \$4 gained in Round 8

**\*\*Key observations:\*\***

- The last sender to me (Round 9) sent nothing
- The receiver in Round 8 returned nothing when I sent \$1
- Two consecutive rounds where cooperation has completely broken down
- Zero reciprocity or trust in the environment

**\*\*Analysis:\*\***

The cooperation level has collapsed to zero. Sending money now means:

- Certain loss of the amount sent
- No expected return based on recent behavior
- Contributing to a non-cooperative equilibrium

The rational decision in this zero-trust environment is to minimize losses.

```
{'send': 0}
```

**Reasoning:**

Let me analyze the situation:

**\*\*Recent pattern:\*\***

- Round 6: I sent \$1.25 as sender
- Round 7: I received \$0 as receiver (sender sent nothing)
- Round 8: I sent \$1.00 as sender
- Round 9: I received \$0 as receiver (sender sent nothing)
- Current earnings: \$21.04

**\*\*Key observations:\*\***

1. Two consecutive rounds as receiver, I got \$0 from senders
2. This shows a very low-trust environment
3. My earnings went from \$17.04 to \$21.04, meaning I gained \$4 in Round 8
  - I earned \$5 - \$1 (sent) + \$0 (returned) = \$4
  - This confirms the receiver in Round 8 returned nothing

**\*\*Pattern analysis:\*\***

- When I send money, I get nothing back (Round 8: sent \$1, got \$0 back)
- When I receive as receiver, senders send nothing (Rounds 7 and 9)
- The cooperation level is essentially zero now

**\*\*Strategy for Round 10:\*\***

Given this completely non-cooperative environment where:

- Senders aren't sending to me
- Receivers aren't returning to me
- There's no reciprocity or trust

The rational choice is to stop sending. Sending anything at this point is just losing money with no expected return.

I should send \$0.

```
Usage: {
 "input_tokens": 720,
 "output_tokens": 544,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:07:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
}
```

```
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$15.990000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0, so I received \$0. Since I received nothing, I have nothing to return. I must return \$0.

**usage:**

```
{
 "input_tokens": 561,
 "output_tokens": 52,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$15.990000000000002.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

```
'''json
{"send": 2}
'''
```

**reasoning:**

Let me think about this situation:

Round 7: I was the sender and sent \$3

Round 8: I was the receiver and the sender sent me \$0, so I received \$0 and returned \$0

Round 9: I'm the sender again

My total earnings are \$15.99 after 8 rounds.

In round 7, I sent \$3. I should have kept \$2 and received something back from the receiver . Looking at my earnings going from \$13.99 to \$15.99, that's an increase of \$2.00.

If I sent \$3 in round 7:

- I kept:  $\$5 - \$3 = \$2$
- The receiver got:  $\$3 \times 3 = \$9$
- If the receiver returned \$0, my earnings would be \$2
- But my earnings increased by \$2, which matches if the receiver returned \$0

Then in round 8:

- I received \$0 (sender sent \$0)
- I returned \$0
- My earnings stayed the same at \$15.99

Now in round 9, I'm the sender again.

The pattern so far suggests:

- In round 7, I sent \$3 and got nothing back
- In round 8, someone sent me \$0

This is interesting because there's mention of communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

Given the game dynamics and that cooperation hasn't been working well (I sent \$3 in round 7 but got nothing back), I might want to be more cautious. However, the game is about trust and reciprocity.

Let me consider my strategy:

1. If I send 0, I keep \$5, the receiver gets \$0, and I can't get anything back
2. If I send some amount, there's a chance for mutual benefit, but also risk

Given that:

- Average earnings so far are about  $\$16/8 = \$2$  per round

**usage:**

```
{
 "input_tokens": 522,
 "output_tokens": 596,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$19.42.

This round, The sender sent you \$0.91, so you received \$2.73. How much do you return to The sender? (0-2.73)

**Assistant output:**

```
'''json
{"return": 1.2}
'''
```

**Reasoning:**

Let me analyze this situation:

1. I'm the RECEIVER in Round 10
2. The sender sent me \$0.91, which was multiplied by 3 to give me \$2.73
3. I need to decide how much to return (between \$0 and \$2.73)

Important context:

- There's communication noise in the game, so amounts may differ from what was actually sent
- My total earnings so far are \$19.42
- The sender sent a relatively small amount (\$0.91)

Looking at the pattern:

- Round 8: I was receiver, sender sent \$0, I returned \$0 (appropriate since I received nothing)
- Round 9: I was sender, I sent \$2
- Round 10: I'm receiver again, sender sent \$0.91 (received \$2.73)

The sender in Round 10 sent a fairly small amount (\$0.91). This could indicate:

1. They're being cautious
2. They don't trust me much
3. They're trying to preserve their earnings
4. There might be noise affecting what they intended to send

A fair return would consider:

- The sender gave up \$0.91 from their \$5
- I received \$2.73
- A reasonable split might be to return about 1/3 to 1/2 of what I received
- This would give the sender a decent return while I still benefit

If I return \$1.00-\$1.50:

- Sender would get:  $\$5 - \$0.91 + \$1.00 - \$1.50 = \$5.09 - \$5.59$
- I would keep:  $\$2.73 - \$1.00 - \$1.50 = \$1.23 - \$1.73$

This seems like a reasonable cooperative response. Given the sender sent less than half of what I sent in Round 9 (\$2), I'll be proportionally generous but not overly so.

I'll return \$1.20, which is about 44% of what I received, giving both of us a decent profit.

Usage: {

```
"input_tokens": 538,
"output_tokens": 494,
"reasoning_tokens": 0
```

}